Multiple Sequence Alignment (MSA) Report Using Clustal Omega

1. Objective

To perform multiple sequence alignment (MSA) of myoglobin proteins from different vertebrate species using Clustal Omega, identify conserved regions and motifs, and study their evolutionary relationships using guide tree and phylogenetic tree analysis.

2. Protein Sequences Used

- Five myoglobin protein sequences were selected for multiple sequence alignment.
- The sequences were retrieved from the UniProt database.
- The selected proteins belong to the same myoglobin protein family.
- The protein sequences used were:
  - Homo sapiens – UniProt ID: P02144 (154 aa)
  - Sus scrofa – UniProt ID: P02189 (154 aa)
  - Mus musculus – UniProt ID: P04247 (154 aa)
  - Rattus norvegicus – UniProt ID: Q9QZ76 (154 aa)
  - Latimeria chalumnae – UniProt ID: H2ZWU1 (154 aa)

3. Methodology

The FASTA sequences of the selected proteins were downloaded from UniProt and submitted to the Clustal Omega tool available at EMBL-EBI. The generated multiple sequence alignment, guide tree, and phylogenetic tree were examined to identify conserved amino acid residues, sequence variations, and evolutionary relationships among the proteins.

4. Conserved Regions and Motifs

The alignment identified several highly conserved regions shared mainly by the selected myoglobin sequences:

- MGLSDGEWQLVLNVWGK
- LIRLFKGH
- DLKKHG...TLALGTILKKKG
- HATKHKIPVKYLEFISE
- SGDFGAD...GAMSKALEL...KELGFQG

The alignment contained numerous "*" symbols, indicating identical amino acids across all aligned sequences. The ":" symbols represented conserved substitutions, while "." symbols indicated semi-conserved substitutions. Gaps represented by "-" indicate positions where residues are absent relative to another sequence. These conserved regions are important for maintaining the characteristic structure and functional properties of myoglobin.

5. Alignment Results and Interpretation

The multiple sequence alignment showed a high degree of sequence similarity among the selected myoglobin proteins. Human, pig, mouse, and rat sequences were particularly similar, with many identical amino acid positions and only a limited number of substitutions. The alignment therefore indicates strong conservation of the myoglobin protein sequence.

The mouse and rat sequences showed a close evolutionary relationship in the phylogenetic tree and were grouped together. Human and pig were also positioned within the mammalian group, showing their evolutionary relationship with the other mammalian myoglobins. The Latimeria chalumnae sequence appeared more distant from the mammalian sequences, indicating greater evolutionary divergence.

The guide tree represents the sequence-grouping pattern used during the progressive alignment, while the phylogenetic tree displays the relative relationships inferred from the aligned sequences. Shorter branches indicate greater sequence similarity, whereas longer branches indicate greater sequence divergence. The observed tree pattern is consistent with the close similarity of the mammalian myoglobin sequences and the more distant position of the coelacanth sequence.

The strong conservation of amino acid residues across the alignment suggests that important structural and functional features of myoglobin have been maintained during evolution. The conserved sequence regions are associated with preservation of the globin fold and the functional environment required for heme and oxygen binding. Species-specific substitutions occur, but the major sequence pattern remains highly conserved.

6. Conclusion

- The Clustal Omega alignment showed that myoglobin is highly conserved among the selected vertebrate species.
- Human, pig, mouse, and rat myoglobin sequences showed very high sequence similarity.
- Mouse and rat formed a closely related group in the phylogenetic analysis.
- Latimeria chalumnae was more evolutionarily distant from the mammalian myoglobins.
- The conserved regions and motifs indicate strong structural and functional conservation of myoglobin.
- Conserved amino acid residues are important for maintaining the protein structure and heme/oxygen-binding function.
- Overall, the analysis demonstrates that myoglobin has been strongly conserved during evolution because of its important biological role.